

sTMS Interface Functional Specification

Singapore Polytechnic – TV Display

Version 1.0

Document Version

Version	Effective Date	Summary of Changes	Created/ Updated By
0.1	10/07/2022	The initial version of sTMS – SP TV Display interface functional specification.	AvePoint
0.2	14/09/2022	Add expiry date to stop display announcement.	AvePoint
0.3	16/09/2022	Change module name to course name	AvePoint
1.0	16/09/2022	Official version for acceptance and sign-off	AvePoint

Document Acceptance Form

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1. Introduction

1.1 Purpose of this document

This Interface Functional Specification (IFS) is to provide the agreed baseline of communication requirements and detailed technical specification between the SaaS shared Training Management System (sTMS) and SP TV Display system for publishing announcement and course schedules that are conducted at T1A building and generate excel file for the user.

New pages in sTMS to enter announcements that will be interfaced with the SP TV.

1.2 Intended Audience

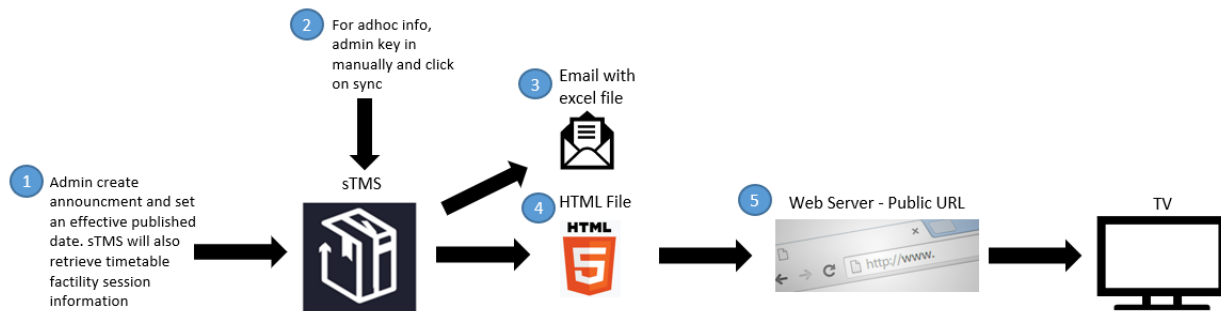
This technical document is used for system-to-system communication. Therefore, the intended audiences are the technical stakeholders of both systems.

2. Technical Overview

2.1 System Flow

There are 3 roles involved in the interface diagram.

- **SP TV:** Singapore Polytechnic T1A Building TV
- **SP Public URL:** A public URL for transferring announcements between sTMS and SP T1A TV
- **sTMS:** shared Training Management System.



SN	Description
1	SP CET school admin create announcements and set an effective published date in sTMS.
2	SP CET school admin is able to create ad-hoc announcements in sTMS and trigger interface to TV.
3	sTMS generates course schedule in excel format and send email to a specific group.
4	sTMS generates course schedule that is conducted at T1A building and announcement on today's date in HTML format.
5	sTMS push HTML file to one web server and then the corresponding information can be displayed on SP TV.

2.2 Interface Design Summary

Output Interface File

Interface File Name	SPTVDISPLAY_{ddMMyyyy}.html
Direction	sTMS to SP TV
Mode of Interface	HTML file
Method	Web Server
Encryption Method	NA
Data Syncing Mode	Full data on today's date
Frequency	Daily
Timing	Scheduled 01:00 AM, for ad-hoc announcements will be manually triggered by SP school admin.
File Transfer Policy	HTML file shall be generated in daily job even if there is no session and announcement need to be published.
Possible Exception and Exception Handling	File transfer failure due to sTMS internal exception such as timer service down. Exception handling steps: • sTMS support team shall be informed by email of the transfer failure. • AP team checks the relevant logs to troubleshoot the cause of the failure. ○ If the issue is due to the transfer files could not be found in SFTP/Web Server, AP redeposits the files in SFTP/Web Server.

Output Interface File

Interface File Name	PACE Academy Daily Automated Classroom Scheduling Report_{ddMMyyyy}.xlsx
Direction	sTMS to SP
Mode of Interface	XLSX
Method	EMAIL
Data Syncing Mode	Full data on today's date
Frequency	Daily
Timing	Scheduled 01:00 AM, for ad-hoc email will be manually triggered by SP school admin in TV announcements page.
File Transfer Policy	Empty XLSX file with header info shall be generated and email to a specific group if there is no session on a certain day.
Possible Exception and Exception Handling	File transfer failure due to sTMS internal exception such as timer service down.

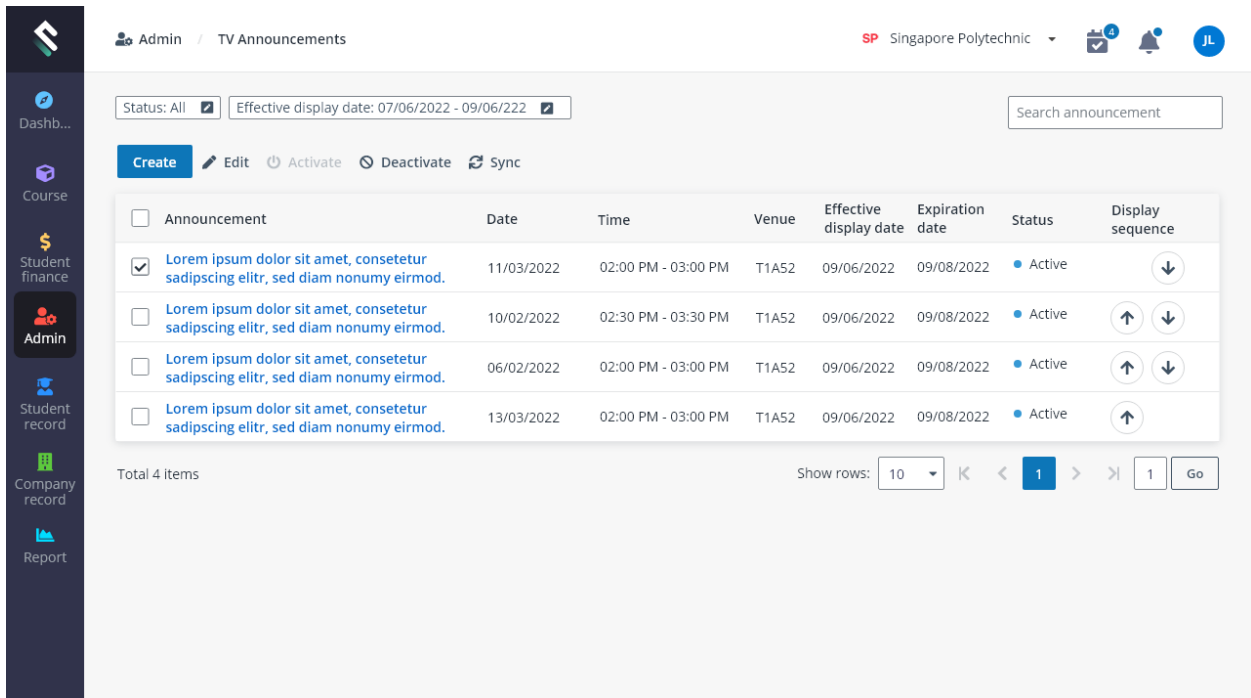
	Exception handling steps: • sTMS support team shall be informed by email of the transfer failure. • AP team checks the relevant logs to troubleshoot the cause of the failure.
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2.3 Interface Process

SP CET school admin is able to create, edit, active, and deactivate TV announcements on this page.

1. Portal navigation: Singapore Polytechnic > Admin > TV Announcements.
2. Only the Announcement that Effective display date = today's date, Venue = T1A building and Status = Active will be sent to SP TV.
3. Once the system generates the file and sends to the TV, an email to be triggered with excel file as an attachment, the email receiver shall be a specific email group.
4. SP admin can trigger the ad-hoc process by Sync button, process needs to trigger the email again with the updated excel file.
5. The excel file will list down all the sessions only for today's date and has not past expiry date regardless of where the session was conducted, user will do the filter on the file on venue if required.
6. Add expiry date to stop displaying announcement

TV Announcements Layouts:



Admin / TV Announcements

Status: All Effective display date: 07/06/2022 - 09/06/2022 Search announcement

Create Edit Activate Deactivate Sync

☐	Announcement	Date	Time	Venue	Effective display date	Expiration date	Status	Display sequence
☒	Lorem ipsum dolor sit amet, consetetur sadipscing elitr, sed diam nonumy eirmod.	11/03/2022	02:00 PM - 03:00 PM	T1A52	09/06/2022	09/08/2022	Active	↓
☐	Lorem ipsum dolor sit amet, consetetur sadipscing elitr, sed diam nonumy eirmod.	10/02/2022	02:30 PM - 03:30 PM	T1A52	09/06/2022	09/08/2022	Active	↑ ↓
☐	Lorem ipsum dolor sit amet, consetetur sadipscing elitr, sed diam nonumy eirmod.	06/02/2022	02:00 PM - 03:00 PM	T1A52	09/06/2022	09/08/2022	Active	↑ ↓
☐	Lorem ipsum dolor sit amet, consetetur sadipscing elitr, sed diam nonumy eirmod.	13/03/2022	02:00 PM - 03:00 PM	T1A52	09/06/2022	09/08/2022	Active	↑

Total 4 items Show rows: 10 1 1 Go

Create Announcement Layout:

Create announcement

Announcement *

Date

01/05/2022

Time ⓘ

Format: hh:mm AM/PM (02:00 PM)

-

Venue

TV settings

Effective display date

01/05/2022

Expiration date

01/08/2022

Cancel

Preview

Save

3. Interface Fields

3.1 TV Announcement Fields

Name	Description	Vaules
S/N	Display Sequence, Running number.	Format: 1, 2, 3
ANNOUNCEMENTS	Announcement in TV.	Format: Long
DATE	Date of session	Format: <i>dd-MM-yy</i>
TIMEFROM	Session start time	Format: <i>hh.mm AM/PM (07:00 PM)</i>
TIMETO	Session end time	Format: <i>hh.mm AM/PM (08:00 PM)</i>
VENUE	Facility ID	Sample: T1A54

3.2 TV Course Schedule Fields

Name	Description	Values
S/N	Sequence number	Format: 1, 2, 3
SESSION	AM/PM	Value list: AM/PM
COURSE/PROGRAMME	sTMS Course name	Format: String
VENUE	Facility ID	Sample: T1A44
DATE	Date of session	Format: <i>dd-MM-yy</i>
TIMEFROM	Session start time	Format: <i>hh.mm AM/PM (07:00 PM)</i>
TIMETO	Session end time	Format: <i>hh.mm AM/PM (08:00 PM)</i>

3.3 Generate HTML File

Course Schedule will be displayed as per template and if schedule has more than 8 rows of data, HTML should be able to show all the data in slide show manner for every 8 rows of data. The announcement page will be shown last after all the schedule displayed and system should be able to go to first page once it's done.

HTML template:

A	B	C	D	E	F
[Logo]	COURSE SCHEDULE FOR [Run Control Date] ([Day])				
S/N	SESSION	COURSE/PROGRAMME	VENUE	TIME	
1	AM	(MEO 4 SL) Re-Sit Exam	T1A32	09:00 AM - 10:40 AM	
2	AM	DNA-PHARMACEUTICALS / BIOLOGICS 2017	T1A34	09:00 AM - 01:00 PM	
3	PM	COC Class 1 & 2 Engineers Part A (Evening Class)	T1A65	06:00 PM - 09:00 PM	
4	PM	PPCDL Theory Test	T1A54	06:00 PM - 09:00 PM	
[Logo]	ANNOUNCEMENT				
S/N	DESCRIPTION		DATE	TIME	VENUE
1	Powered Pleasure Craft Driving Licence (PPCDL) Theory Test venue has changed from T1A62 to T1A54. Please report to the test venue early.		11/7/2022	07:00 PM - 08:00 PM	T1A54



SPTVDISPLAY_11072
022.html

3.4 Classroom scheduling excel file fields

Name	Description	Values
S/N	Sequence number	Sample: 1, 2, 3
Course ID	sTMS Course code	
Run Number	sTMS class name	Sample: class 1, class2
Course START Date	CourseStartDate Semester based course - SemesterStartDate	Format: dd-MM-yyyy
Course END Date	CourseEndDate Semester based course - SemesterEndDate	Format: dd-MM-yyyy
Course Name	sTMS Course name	Format: String
Lecturer Name	Lecturer of session, if instructor is more than one, display all name.	Sample: Qwert, Yuiop
Start Time	Session start time	Format: hh.mm (9:00)
End Time	Session end time	Format: hh.mm (13:00)

Venue	Facility ID	Sample: T1A26
PA in Charge	sTMS course manager name	Format: String
No. of Pax in Classlist	Number of students in class	Format: number
Run Control Date	Session date = today's date	Format: dd-MM-yyyy

3.5 Generate Email and Excel File

1. Email Details:

- To:** a specific email group
- Subject:** PACE Academy Daily Automated Classroom Scheduling Report for <Process Date>, <Process Time>
- Body:**
 - REPORT DETAILS -

Report Title: PACE Academy Daily Automated Classroom Scheduling Report

Date Generated: <Process Date>

Time Generated: <Process Time>

Generated by: <sTMS>

Mode: <Batch / Ad Hoc>

- INSTRUCTIONS TO CS TEAM -

(Step-1) Review the scheduling details in the attached report.

(Step-2) If there is any missing venue information in the report, follow up with the PA concerned to ensure that venue details are properly updated in sTMS.

(Step-3) Generate another report to ensure that updates and details are properly reflected.

(Step-4) Inspect to ensure that the scheduling details published (e.g. on the TV etc.) are also updated accordingly.

(Step-5) Share the most updated report to all frontline staff for their reference.

- Attachment (excel) File Name:** PACE Academy Daily Automated Classroom Scheduling Report_{ddMMyyyy}.xlsx

2. Attachment: Excel file.

	A	B	C	D	E	F	G	H	I	J	K	L
1						PACE ACADEMY						
2						DAILY AUTOMATED CLASSROOM SCHEDULING REPORT						
3						FOR [Run Control Date]						
4						GENERATED ON [Today Date]						
5												
6	S/N	Course ID	Run Number	Course START Date	Course END Date	Course Name	Lecturer Name	Start Time	End Time	Venue	PA in Charge	No. of Pax in Classlist
7	1	500143	Class 1	20/7/2017	28/7/2017	Water Efficiency Manager Course	Asdfg	9:00	12:00	T1A26	Zxcvbnm	12
8	2	500143	Class 2	20/7/2017	28/7/2017	Water Efficiency Manager Course	Qwert, Yulop	13:00	15:00	T1A33	Zxcvbnm	15



PACE Academy Daily
Automated Classroom

Time in 24Hrs format. Sort by Course Start Date, Start Time, Course Name, ascending.